

AI PM Learning Roadmap

[Read the full post](#)

1. Basic Concepts

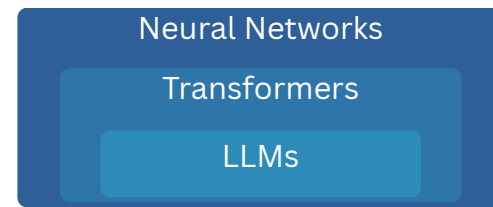
Machine Learning:

- ☐ Supervised Learning
- ☐ Unsupervised Learning
- ☐ Reinforcement Learning
- ☐ Deep Learning

Free resources:

- [WTF is AI Product Manager](#)
- [LLM Visualization](#)

Architectures:



Models:



2. Prompt Engineering

Free guides & courses:

- [GPT-4.1 Prompting Guide](#)
- [Anthropic Prompt Eng.](#)
- [Prompt Eng. by Google](#)
- [System prompt, Claude 4](#)
- [Anthropic Prompt Generator](#)
- [Anthropic Prompt Library](#)
- [Anthropic Prompting Course](#)

Techniques:



3. Fine-Tuning

Methods:

- ☐ Supervised (SFT)
- ☐ Direct Prefer. Opt. (DPO)

Best tools:

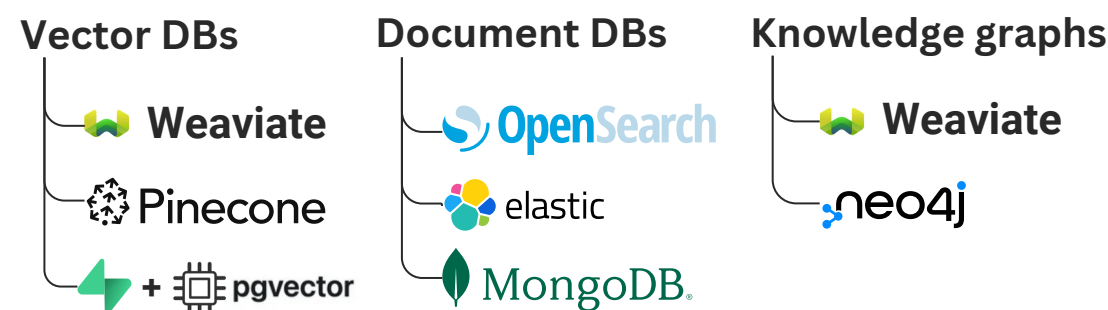
- [OpenAI Platform](#)
- [Hugging Face AutoTrain](#)
- [LLaMA-Factory](#)

Terms:

- ☐ Training data
- ☐ Validation data
- ☐ Epoch
- ☐ Batch size
- ☐ Learning rate
- ☐ Beta (for DPO)

Key Metric	What it Tells You
Training Loss ↓	The model is learning from data.
Training Accuracy ↑	Better at predicting tokens it has seen.
Validation Loss ↓	It's generalizing well to new data.
Validation Accuracy ↑	It's predicting new tokens well.

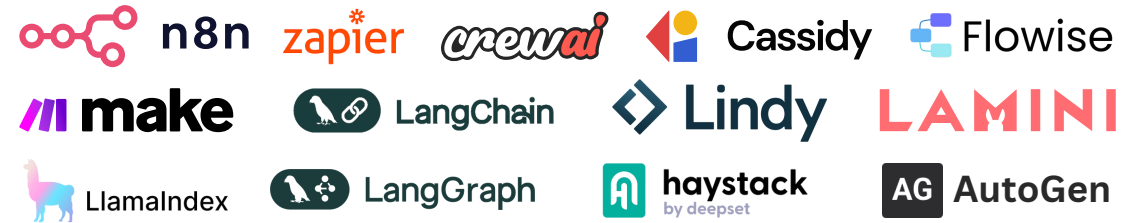
4. RAG (Retrieval-Augmented Gen.)



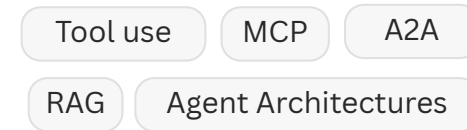
Vector DB vs. Document DB

Use **vector store** to store embeddings and perform fast semantic search over chunks. Use **document DB** to store the full content and metadata (e.g., document url, author, date).

5. AI Agents & Agentic Workflows



Techniques:

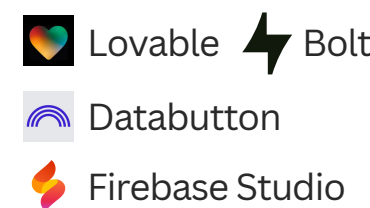


Free reports:

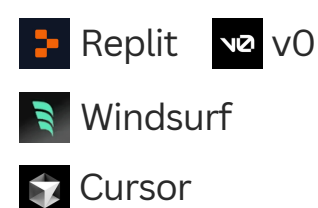
- [Google Agent Companion](#)
- [Anthropic Building Agents](#)
- [IBM Agentic Process Automation](#)

6. AI Prototyping & Building

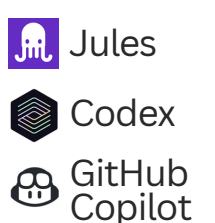
No-code first:



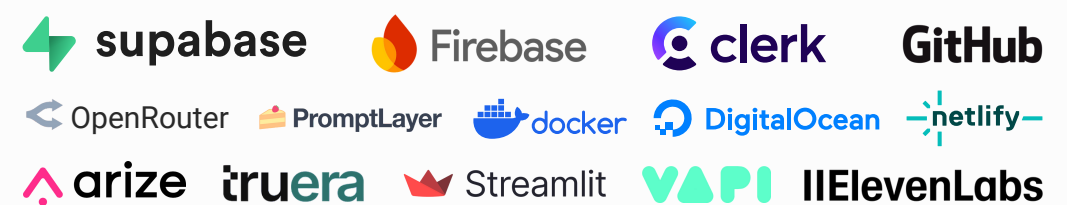
IDE first:



Others:



Infrastructure Tools

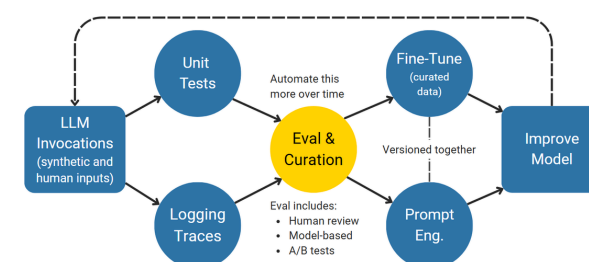


7. Foundational Models

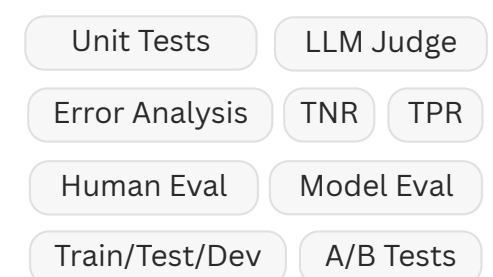


8. AI Evaluation Systems

Virtuous Cycle:



Techniques:



- [Mastering AI Evals: A Free, Complete Guide for PMs](#)

9. Other Resources

- [AI PRD: A free PRD template by Miqdad Jaffer \(OpenAI\)](#)
- [Awesome Generative AI Guide \(GitHub\)](#)
- [ChatLLM \(abacus.ai\): All LLMs in one subscription](#)
- [MCP.so: The largest collection of MCP servers](#)
- [Anthropic MCP Servers: The official Anthropic collection](#)
- [microsoft/markitdown: For converting docs to Markdown](#)